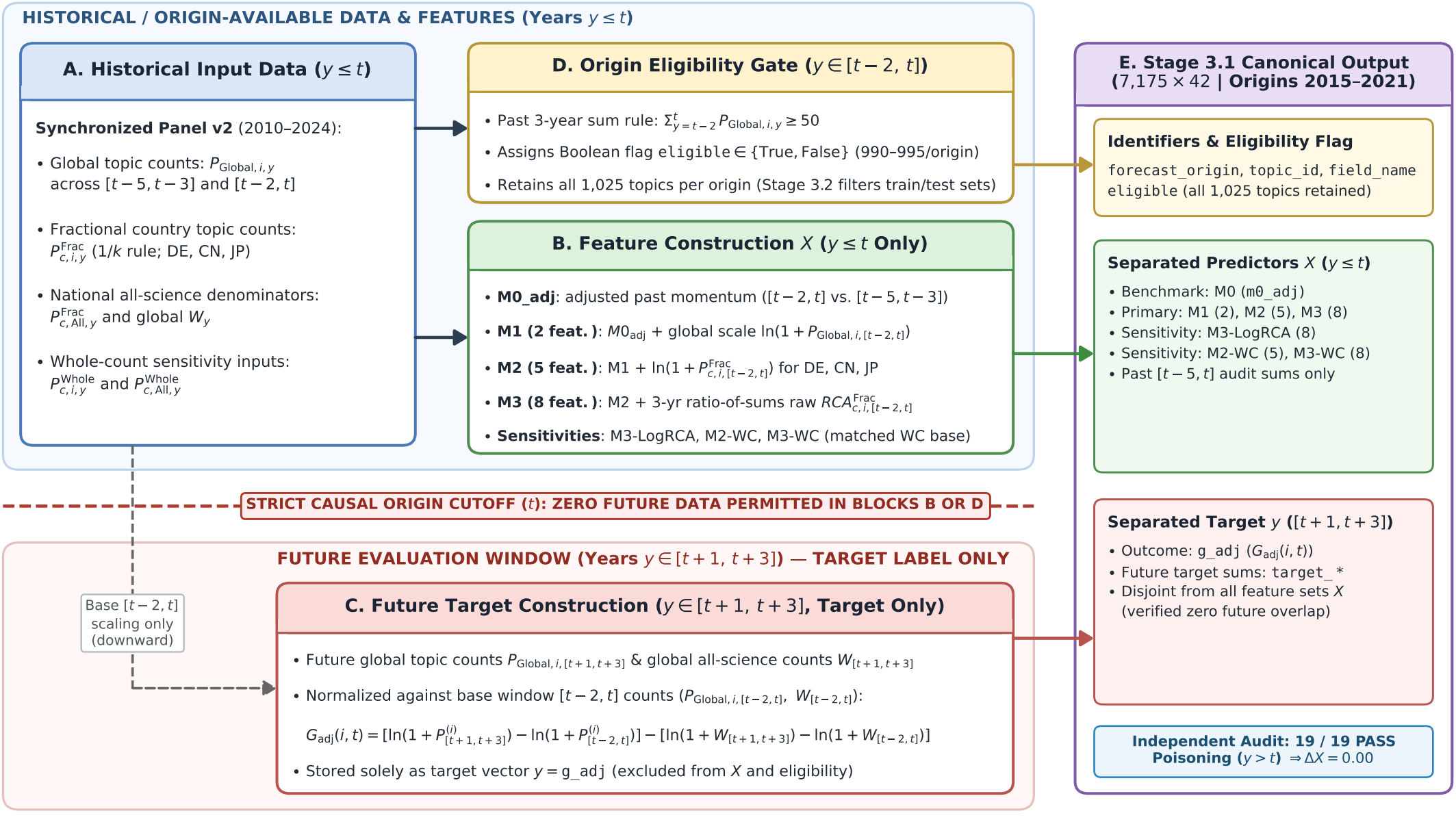

HISTORICAL / ORIGIN-AVAILABLE DATA & FEATURES (Years $y \le t$)
A. Historical Input Data ($y \le t$)
Synchronized Panel v2 (2010–2024):
• Global topic counts: $P_{\text{Global},i,y}$ across $[t-5, t-3]$ and $[t-2, t]$
• Fractional country topic counts: $P^{\text{Frac}}_{c,i,y}$ (1/k rule; DE, CN, JP)
• National all-science denominators: $P^{\text{Frac}}_{c,\text{All},y}$ and global W_y
• Whole-count sensitivity inputs: $P^{\text{Whole}}_{c,i,y}$ and $P^{\text{Whole}}_{c,\text{All},y}$
D. Origin Eligibility Gate ($y \in [t-2, t]$)
• Past 3-year sum rule: $\Sigma^{t}_{y=t-2} P_{\text{Global},i,y} \ge 50$
• Assigns Boolean flag eligible ∈ {True, False} (990–995/origin)
• Retains all 1,025 topics per origin (Stage 3.2 filters train/test sets)
B. Feature Construction X ($y \le t$ Only)
• M0_adj: adjusted past momentum ($[t-2, t]$ vs. $[t-5, t-3]$)
• M1 (2 feat.): $M0_{\text{adj}}$ + global scale $\ln(1 + P_{\text{Global},i,[t-2,t]})$
• M2 (5 feat.): M1 + $\ln(1 + P^{\text{Frac}}_{c,i,[t-2,t]})$ for DE, CN, JP
• M3 (8 feat.): M2 + 3-yr ratio-of-sums raw $RCA^{\text{Frac}}_{c,i,[t-2,t]}$
• Sensitivities: M3-LogRCA, M2-WC, M3-WC (matched WC base)
STRICT CAUSAL ORIGIN CUTOFF (t): ZERO FUTURE DATA PERMITTED IN BLOCKS B OR D
FUTURE EVALUATION WINDOW (Years $y \in [t+1, t+3]$) — TARGET LABEL ONLY
Base $[t-2, t]$ scaling only (downward)
C. Future Target Construction ($y \in [t+1, t+3]$, Target Only)
• Future global topic counts $P_{\text{Global},i,[t+1,t+3]}$ & global all-science counts $W_{[t+1,t+3]}$
• Normalized against base window $[t-2, t]$ counts ($P_{\text{Global},i,[t-2,t]}$, $W_{[t-2,t]}$):
$G_{\text{adj}}(i,t) = [\ln(1 + P^{(i)}_{[t+1,t+3]}) - \ln(1 + P^{(i)}_{[t-2,t]})] - [\ln(1 + W_{[t+1,t+3]}) - \ln(1 + W_{[t-2,t]})]$
• Stored solely as target vector y = g_adj (excluded from X and eligibility)
E. Stage 3.1 Canonical Output (7,175 × 42 | Origins 2015–2021)
Identifiers & Eligibility Flag
forecast_origin, topic_id, field_name
eligible (all 1,025 topics retained)
Separated Predictors X ($y \le t$)
• Benchmark: M0 (m0_adj)
• Primary: M1 (2), M2 (5), M3 (8)
• Sensitivity: M3-LogRCA (8)
• Sensitivity: M2-WC (5), M3-WC (8)
• Past $[t-5, t]$ audit sums only
Separated Target y ($[t+1, t+3]$)
• Outcome: g_adj ($G_{\text{adj}}(i,t)$)
• Future target sums: target_*
• Disjoint from all feature sets X (verified zero future overlap)
Independent Audit: 19 / 19 PASS
Poisoning ($y > t$) ⇒ $\Delta X = 0.00$